24CS2016 Flutter Lab - Final Cheat Sheet

Ready-to-Use Terminal Commands & Implementation Code Snippets

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Target: Android & Web Build

Exp 1: Working With Widgets and Layouts

CORE UI

TERMINAL COMMANDS

```
flutter create widget_layout && cd widget_layout && code . && flutter run
```

COPY-PASTE SNIPPET (INSIDE BODY: COLUMN)

```
Card(
  elevation: 4,
  child: ListTile(
    leading: const Icon(Icons.book, size: 35),
    title: const Text('Flutter', style: TextStyle(fontWeight: FontWeight.bold)),
    subtitle: const Text('Mobile Application Development'),
    trailing: const Icon(Icons.arrow_forward_ios),
  ),
)
```

Exp 2: Adaptive & Responsive UI Design

LAYOUTBUILDER

TERMINAL COMMANDS

```
flutter create responsive_ui && cd responsive_ui && code . && flutter run
```

COPY-PASTE SNIPPET (RESPONSIVE LAYOUT ENGINE)

```
LayoutBuilder(
  builder: (context, constraints) {
    if (constraints.maxWidth < 600) {
      return Column(children: const [Card(child: ListTile(title: Text('Student
Mobile')))]);
    } else {
      return Row(children: const [Expanded(child: Card(child: ListTile(title:
Text('Student Desktop'))))]);
    }
  },
)
```

Exp 3: Interactivity - Forms & Validation

FORM STATE

TERMINAL COMMANDS

```
flutter create student_form && cd student_form && code . && flutter run
```

COPY-PASTE SNIPPET (FORM KEY + TEXTFORMFIELD + SUBMIT BUTTON)

```
final _formKey = GlobalKey<FormState>();

Form(
  key: _formKey,
  child: Column(children: [
    TextFormField(
      decoration: const InputDecoration(labelText: 'Student Name'),
      validator: (v) => (v == null || v.isEmpty) ? 'Enter student name' : null,
    ),
    ElevatedButton(
      onPressed: () {
        if (_formKey.currentState!.validate()) {
          ScaffoldMessenger.of(context).showSnackBar(const SnackBar(content:
Text('Success')));
        }
      },
      child: const Text('Submit'),
    ),
  ]),
)
```

Exp 4: Media Assets & GridView

PUBSPEC.YAML

TERMINAL COMMANDS

```
flutter create media_app && cd media_app && mkdir -p assets/images && flutter pub get
```

PUBSPEC.YAML ADDITION

```
flutter:
  assets:
    - assets/images/
```

COPY-PASTE SNIPPET (GRIDVIEW GALLERY)

```
GridView.count(
  crossAxisCount: 2,
  children: [
    Image.asset('assets/images/flutter.png'),
  ],
)
```

Exp 5: Navigation & Routing

NAVIGATOR

TERMINAL COMMANDS

```
flutter create navigation_app && cd navigation_app && code . && flutter run
```

COPY-PASTE SNIPPET (NAMED ROUTES & ACTIONS)

```
// In MaterialApp:
routes: {
  '/': (context) => const HomePage(),
  '/profile': (context) => const ProfilePage(),
},

// Navigation Actions:
Navigator.pushNamed(context, '/profile'); // Push route
Navigator.pop(context);                   // Return back
```

Exp 6: Implicit Animations & Transitions

ANIMATEDCONTAINER

TERMINAL COMMANDS

```
flutter create animation_app && cd animation_app && code . && flutter run
```

COPY-PASTE SNIPPET (STATE VARIABLE + ANIMATION)

```
bool isLarge = false;

AnimatedContainer(
  duration: const Duration(seconds: 1),
  width: isLarge ? 300 : 150,
  height: isLarge ? 300 : 150,
  child: const Center(child: Text('Flutter')),
);

// Inside Button Callback:
setState(() { isLarge = !isLarge; });
```

Exp 7: State Management & JSON Serialization

MODEL & LISTVIEW

TERMINAL COMMANDS

```
flutter create state_app && cd state_app && code . && flutter run
```

COPY-PASTE SNIPPET (STUDENT MODEL + SERIALIZER)

```
class Student {
  final int id;
  final String name;
  Student({required this.id, required this.name});

  Map<String, dynamic> toJson() => {'id': id, 'name': name};
  factory Student.fromJson(Map<String, dynamic> json) => Student(id: json['id'], name:
json['name']);
}

// Display List:
ListView.builder(
  itemCount: students.length,
  itemBuilder: (context, index) => ListTile(title: Text(students[index].name)),
)
```

Exp 8: Data Persistence (SQLite & Firebase)

SQFLITE

TERMINAL COMMANDS

```
flutter create sqlite_app && cd sqlite_app && flutter pub add sqflite path
```

COPY-PASTE SNIPPET (SQLITE DATABASE CREATION)

```
import 'package:sqflite/sqflite.dart';
import 'package:path/path.dart';

Future<Database> initDB() async {
  return openDatabase(
    join(await getDatabasesPath(), 'student.db'),
    onCreate: (db, version) {
      return db.execute('CREATE TABLE students(id INTEGER PRIMARY KEY, name TEXT, course
TEXT)');
    },
    version: 1,
  );
}
```

Exp 9: Accessing APIs through HTTP

REST API

TERMINAL COMMANDS

```
flutter create api_app && cd api_app && flutter pub add http
```

COPY-PASTE SNIPPET (HTTP GET REQUEST)

```
import 'dart:convert';
import 'package:http/http.dart' as http;

Future<void> fetchData() async {
  final res = await http.get(Uri.parse('https://jsonplaceholder.typicode.com/posts'));
  if (res.statusCode == 200) {
    List data = jsonDecode(res.body);
    print('Fetched ${data.length} records');
  } else {
    print('Failed to load data: ${res.statusCode}');
  }
}
```

Exp 10: Add Android as a Build Target

APK BUILD

TERMINAL COMMANDS (STEP-BY-STEP SEQUENCE)

```
flutter doctor --android-licenses
```

```
flutter create android_app && cd android_app
```

```
flutter emulators --launch Medium_Phone_API_36.1
```

```
flutter run
```

```
flutter build apk
```

OUTPUT BINARY PATH

```
build/app/outputs/flutter-apk/app-release.apk
```